EKACTL-Typst, section 2

Biochemistry

Note that when we use log in this section, we mean $\log_{10}$.

1. pH and buffers

1.1. pH and pOH

$[\mathrm{H}^+]$ is the molarity of H^+ in whatever solution we're measuring the pH of.

$$\mathrm{pH} = -\log([\mathrm{H}^+]) \quad [\mathrm{H}^+] = 10^{-\mathrm{pH}}$$

$$\mathrm{pOH} = -\log([\mathrm{OH}^-])[\mathrm{OH}^-] = 10^{-\mathrm{pOH}}$$

$$\mathrm{pH} + \mathrm{pOH} = 14$$

1.2. Strong acids and bases

The following are the strong acids to memorize:

- H_2SO_4 (sulfuric acid)
- HNO_3 (nitric acid)
- HCl (hydrochloric acid)

The following are the strong bases to memorize:

- NaOH (sodium hydroxide)
- KOH (potassium hydroxide)
- $Ca(OH)_2$ (calcium hydroxide)

Definition 1.2.1 (Dissociation of strong acids and bases)

The reaction in which a strong acid or base breaks apart into a proton (H^+) (for a strong acid) or a hydroxide (OH^-) (for a strong base) and another ion.

Strong acids and bases dissociate completely in water.

1.3. Protonation and ionization

Definition 1.3.1 (Protonation and ionization)

Protonation is the gain of a proton.

Ionization is the loss of a proton.

1.4. Weak acids

Definition 1.4.1 (Conjugate base)

The non-proton product of an acid ionization reaction in which an acid loses its proton.

Definition 1.4.2 (Ionization of a weak acid)

The reaction in which part of a weak acid breaks apart into a proton and its conjugate base. The conjugate base is often written as A^-.

$$\mathrm{HA(aq)} \rightleftharpoons \mathrm{H}^{+(\mathrm{aq})} + \mathrm{A}^{-(\mathrm{aq})}$$

This often occurs in water. Note that only a small amount of most weak acids ionize.

Definition 1.4.3 (Equilibrium constant for ionization of a weak acid)

$$K_a = \frac{[\mathrm{H^+}][\mathrm{A^-}]}{[\mathrm{HA}]}$$

The negative log of K_a is called $\mathrm{p}K_a$:

$$\mathrm{p}K_a = -\log K_a$$

Stronger acids have higher K_a and lower $\mathrm{p}K_a$.

The following equation is easily derived from taking the log of both sides of the definition of K_a and rearranging.

Theorem 1.4.4 (Henderson-Hasselbalch equation)

$$\begin{aligned} \mathrm{pH} &= \mathrm{p}K_a + \log\left(\frac{[\mathrm{A^-}]}{[\mathrm{HA}]}\right) \\ &= \mathrm{p}K_a + \log\left(\frac{[\text{conjugate base}]}{[\text{acid}]}\right) \end{aligned}$$

1.5. Buffers

Definition 1.5.1 (Buffer)

A solution that can resist change in pH upon the addition of an acid or a base.

Buffers can be created using a weak acid and its conjugate base, or a weak base and its conjugate acid.

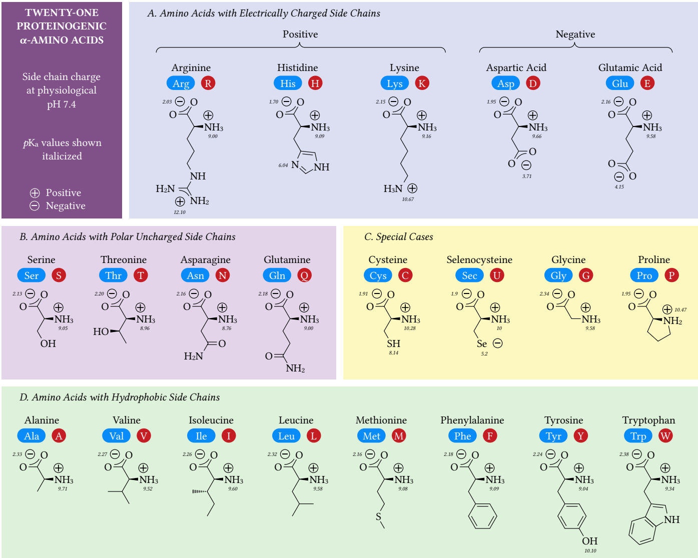
1.5.1. Behavior of weak acid-conjugate base buffers

1.5.2. Bicarbonate buffer system

$$CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$$

2. Amino acids

2.1. Proteinogenic amino acids


TWENTY-ONE PROTEINOGENIC α-AMINO ACIDS
Side chain charge at physiological pH 7.4
pKa values shown italicized
⊕ Positive
⊖ Negative
A. Amino Acids with Electrically Charged Side Chains
Positive
Negative
Arginine Arg R 2.03 9.00 12.10
Histidine His H 1.70 9.09 6.04
Lysine Lys K 2.15 9.16 10.67
Aspartic Acid Asp D 1.95 9.66 3.71
Glutamic Acid Glu E 2.16 9.58 4.15
B. Amino Acids with Polar Uncharged Side Chains
Serine Ser S 2.13 9.05
Threonine Thr T 2.20 8.96
Asparagine Asn N 2.16 8.76
Glutamine Gln Q 2.18 9.00
C. Special Cases
Cysteine Cys C 1.91 10.28 8.14
Selenocysteine Sec U 1.9 10 5.2
Glycine Gly G 2.34 9.58
Proline Pro P 1.95 10.47
D. Amino Acids with Hydrophobic Side Chains
Alanine Ala A 2.33 9.71
Valine Val V 2.27 9.52
Isoleucine Ile I 2.26 9.60
Leucine Leu L 2.32 9.58
Methionine Met M 2.16 9.08
Phenylalanine Phe F 2.18 9.09
Tyrosine Tyr Y 2.24 9.04 10.10
Tryptophan Trp W 2.38 9.34

2.2. Types of amino acids

Definition 2.2.1 (Neutral nonpolar amino acid)

An amino acid whose side chain contains mostly hydrocarbons. These are also called hydrophobic amino acids.

Aromatic amino acids: Phenylalanine, tryptophan

Aliphatic amino acids: Glycine, alanine, valine, leucine, isoleucine, proline, methionone, cysteine

Definition 2.2.2 (Neutral polar amino acid)

These are also called hydrophilic amino acids.

Examples: Serine, threonine, tyrosine, asparagine, glutamine

Definition 2.2.3 (Acidic amino acid)

An amino acid whose side chain is negatively charged at pH ≈ 7. All standard acidic amino acids have carboxylate groups in their side chains.

Examples: Aspartic acid, glutamic acid

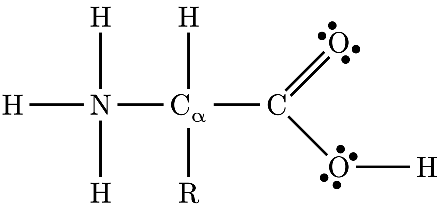
Definition 2.2.4 (Basic amino acid)

An amino acid whose side chain is positively charged at pH ≈ 7.

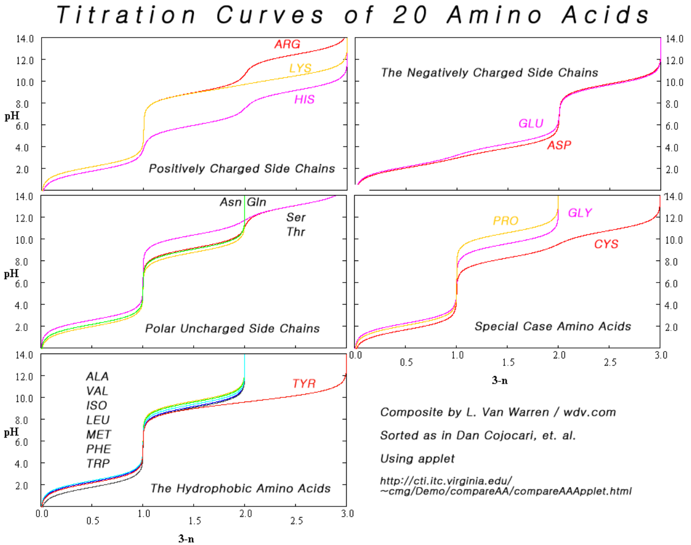
Examples: Lysine, histidine, arginine

2.3. Charges and titration of amino acids

Amino acid at low pH:



As pH increases, first COOH loses a proton (becoming COO^-), and then NH_3^+ loses a proton (becoming NH_2). If the amino acid is acidic or basic, at some pH it will also lose a proton from its side chain.


Titration Curves of 20 Amino Acids

Composite by L. Van Warren / wdv.com

Sorted as in Dan Cojocari, et. al.

Using applet

http://cti.itc.virginia.edu/~cmg/Demo/compareAA/compareAAApplet.html

Definition 2.3.1 (Isoelectric point)

The pH at which the amino acid has no net charge.

It is also denoted **pI**.

Definition 2.3.2 (Zwitterion)

The form of an amino acid which has no net charge.

Lemma 2.3.3

Of the following $\mathrm{p}K_a$s:

- The $\mathrm{p}K_a$ of the amino group ($\mathrm{NH}_3^+ \rightleftharpoons \mathrm{NH}_2$)
- The $\mathrm{p}K_a$ of the carboxyl group ($\mathrm{COO}^- \rightleftharpoons \mathrm{COOH}$)
- The $\mathrm{p}K_a$ of the side chain

The isoelectric point is:

- For acidic (positive) amino acids, the mean of the two lowest $\mathrm{p}K_a$s
- For basic (negative) amino acids, the mean of the two highest $\mathrm{p}K_a$s
- For nonpolar and polar amino acids, the mean of the $\mathrm{p}K_a$ of the amino group and the $\mathrm{p}K_a$ of the carboxyl group (the only two $\mathrm{p}K_a$s)

Theorem 2.3.4

$$\text{Formal charge} = \text{valence electrons} - (\text{dots} + \text{bonds})$$

2.4. Skeletal drawing

Carbons are not shown, and hydrogens bonded to carbons are not shown.

2.5. Redox

Definition 2.5.1 (Oxidation and reduction)

Oxidation is a process in which a substance loses electrons.

Reduction is a process in which a substance gains electrons.

Oxidation and reduction must occur together.

Definition 2.5.2 (Oxidation state)

The hypothetical charge of an atom if all of its bonds to other atoms were fully ionic.

In covalent bonds, we assign the electrons to the more electronegative atom.

Lemma 2.5.3

In carbon-hydrogen bonds, the electron is assigned to the carbon.

In carbon-oxygen bonds, the electron is assigned to the oxygen.

In carbon-carbon bonds, the electrons are shared equally.

Example. The following are half-reactions; the top is a reduction and the bottom is an oxidation.

$$\mathrm{Cu}^{2+} + 2\mathrm{e}^{-} \longrightarrow \mathrm{Cu} \qquad \text{(reduction)}$$
$$\mathrm{Zn} \longrightarrow \mathrm{Zn}^{2+} + 2\mathrm{e}^{-} \qquad \text{(oxidation)}$$

The redox reaction is obtained by adding the half reactions and simplifying:

$$\mathrm{Cu}^{2+} + 2\mathrm{e}^{-} + \mathrm{Zn} \longrightarrow \mathrm{Cu} + \mathrm{Zn}^{2+} + 2\mathrm{e}^{-}$$
$$\mathrm{Cu}^{2+} + \mathrm{Zn} \longrightarrow \mathrm{Cu} + \mathrm{Zn}^{2+}$$

2.6. Protein structure

Definition 2.6.1

Primary structure is the order of the amino acids.

Secondary structure is the α-helices and β-pleated sheets, formed by hydrogen bonds between the peptide bonds.

Tertiary structure is the overall shape of each polypeptide, which is determined by side-chain characteristics such as:

- Nonpolar (hydrophobic) amino acids going towards the center
- Disulfide bonds
  - Disulfide bonds cannot be broken without a

reducing agent
- Salt bridges (ionic bonds)
- Hydrogen bonds on the side chains

Quaternary structure is multiple polypeptide chains together, held together by the same forces that create tertiary structure.

Definition 2.6.2 (Hydrogen bond)

An intermolecular force between a hydrogen bonded to N, O, or F and a lone pair.

2.7. Thin layer chromatography, polarity

Lemma 2.7.1

For amino acids in the same group (same charge and polar/nonpolar), longer and bulkier hydrocarbon side chains increase hydrophobicity and nonpolarity.

Thus, within the same group, one can count the number of carbons; more carbons means more nonpolar.

Theorem 2.7.2 (Beer-Lambert law)

The absorbance of light passing through an object, where the path the light takes through the object has length l, with molar absorptivity ε and concentration c is given by

$$A = \varepsilon l c$$

3. Enzymes

Definition 3.1 (Catalyst)

A substance that increases the rate of a chemical reaction and is not consumed by the reaction.

Definition 3.2 (Enzyme)

A biological macromolecule that is a catalyst.

Definition 3.3 (Substrate)

The reactant in an enzyme-catalyzed reaction.

The substrate binds to the enzyme at the ***active site***.

3.1. Cofactors

Definition 3.1.1 (Cofactor)

Something that the enzyme needs to bind to in order to work.

Definition 3.1.2 (Coenzyme)

A cofactor which is an organic molecule, loosely bound to the enzyme.

Definition 3.1.3 (Prosthetic group)

A cofactor tightly bound to the enzyme, which stays permanently attached.

Definition 3.1.4 (Holoenzyme)

The enzyme with all of its cofactors.

Definition 3.1.5 (Apoenzyme)

The enzyme without its cofactors.

3.2. Types of enzymes

Definition 3.2.1 (Oxidoreductase)

Catalyzes redox reactions.

Definition 3.2.2 (Transferase)

Catalyzes the transfer of a functional group from one molecule to another.

Definition 3.2.3 (Hydrolase)

Catalyzes hydrolysis (breaking bonds by reacting with water)

Definition 3.2.4 (Lyase)

Catalyzes the breaking of bonds without water nor oxidation

Definition 3.2.5 (Isomerase)

Catalyzes the rearrangement of the atoms in the molecule.

Definition 3.2.6 (Ligase)

Catalyzes the joining of molecules.

3.3. Chemical reaction kinetics

Definition 3.3.1 (Rate)

The ***rate*** or ***velocity*** of a chemical reaction with one product is the change in concentration of the product over the change in time.

Given a balanced chemical reaction with two reactants and two products:

$$\mathrm{aA} + \mathrm{bB} \longrightarrow \mathrm{pP} + \mathrm{qQ}$$

the rate is defined as:

$$v = -\frac{1}{a}\frac{d[\mathrm{A}]}{dt} = -\frac{1}{b}\frac{d[\mathrm{B}]}{dt} = \frac{1}{p}\frac{d[\mathrm{P}]}{dt} = \frac{1}{q}\frac{d\mathrm{Q}}{dt}$$

Definition 3.3.2 (Rate constant, rate equation)

The ***rate constant*** k of a reaction is inherent to the reaction and is related to its speed.

Assume a reaction $\mathrm{A} \longrightarrow \mathrm{B}$.

A zero-order reaction has a constant rate:

$$v = k$$

A first-order reaction has a linearly increasing rate:

$$v = k[\mathrm{A}]$$

A second-order reaction has a quadratically increasing rate:

$$v = k[\mathrm{A}]^2$$

3.4. Enzyme kinetics

Definition 3.4.1 (Enzyme-catalyzed reaction)

A reaction:

$$\mathrm{E} + \mathrm{S} \xleftrightarrow[\mathrm{k_3}]{\mathrm{k_1}} \mathrm{ES} \xrightarrow{\mathrm{k_2}} \mathrm{E} + \mathrm{P}$$

E represents the free enzyme, $\mathrm{E_T}$ represents the total enzyme, S represents the substrate, and P represents the product.

Definition 3.4.2 (Michaelis constant)

The ***Michaelis constant*** is defined as:

$$K_m := \frac{k_2 + k_3}{k_1}$$

A low K_m causes a quick reaction, while a high K_m causes a slower reaction.

Definition 3.4.3 (K_{cat})

$$k_{\text{cat}} := \frac{v_{\text{max}}}{[\text{E}]}$$

k_{cat} and K_m are both inherent to each enzyme, and are invariant with respect to concentration, etc.

$\frac{k_{\text{cat}}}{K_m}$ is the best measure of enzyme efficiency.

Theorem 3.4.4 (Michaelis-Menton equation)

The maximum velocity of an enzyme-catalyzed reaction is

$$v_{\text{max}} = k_2[\text{E}_\text{T}]$$

The velocity of an enzyme-catalyzed reaction is given by the ***Michaelis-Menton equation***:

$$v = \frac{v_{\text{max}}\,[\text{S}]}{K_m + [\text{S}]}$$

> The equivalent, linear ***Lineweaver-Burk equation*** is:
>
> $$\frac{1}{v} = \frac{K_m}{v_{\max}\,[\mathrm{S}]} + \frac{1}{v_{\max}}$$
>
> Note also that the x-intercept of $\frac{1}{[\mathrm{S}]}$ is equal to $-\frac{1}{K_m}$.

3.5. Enzyme regulation

There are three types of enzyme regulation:

- genetic regulation
- covalent modification (phosphorylation)
- allosteric regulation

3.6. Enzyme inhibition

3.6.1. Reversible inhibitors